Systems Thinking

A Short Guide

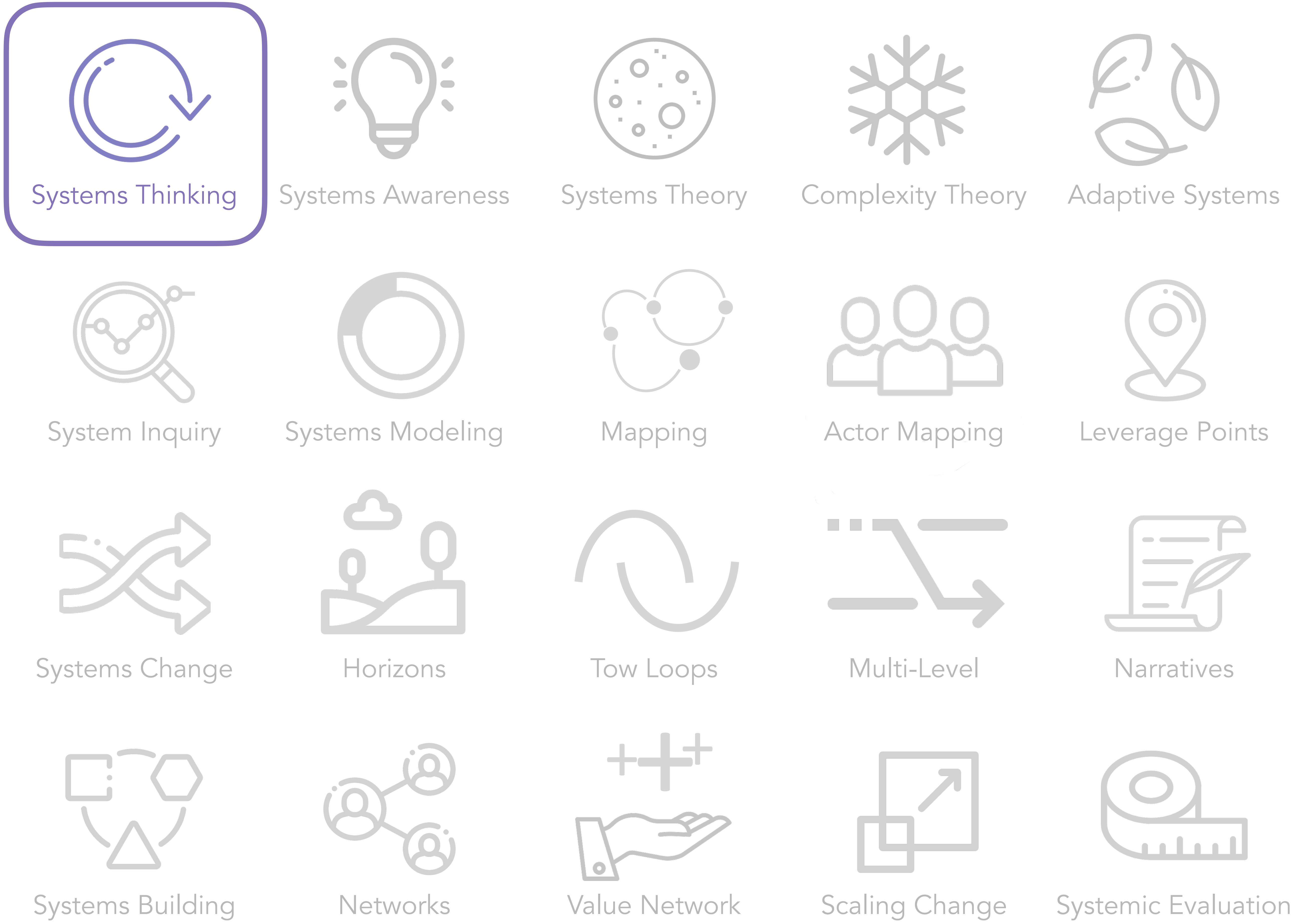
Overview

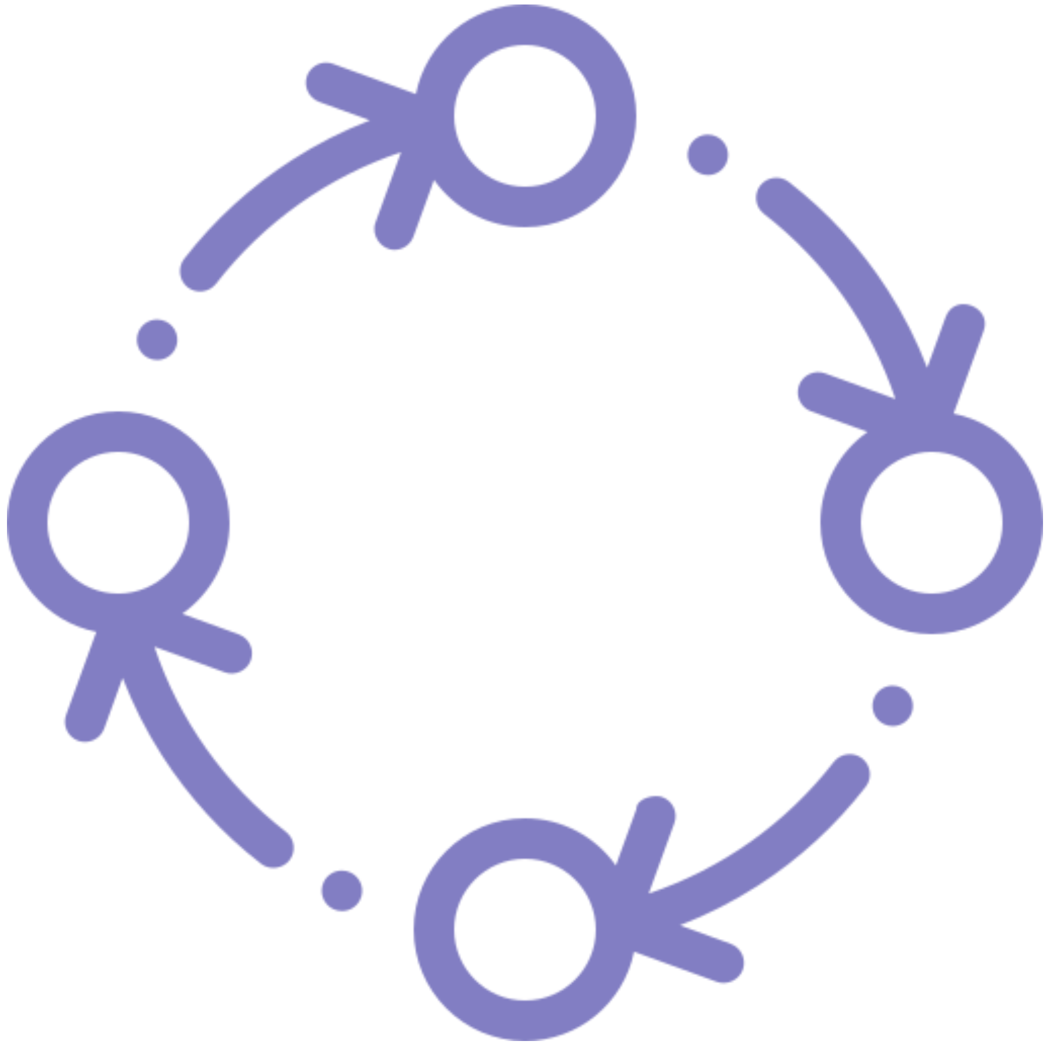
This guide is designed for those teaching or facilitating people to learn and apply systems thinking. It has been created particularly for those involved in the practice of systems change or Systems Innovation Labs.

The guide gives an outline to what systems thinking is and the different aspects that constitute it. This is done in a compact and succinct form so that facilitators can use it as a foundation to scaffold their training. As such the contents of this guide should be complemented and fleshed out by the facilitator with their additional material; worksheets, quizzes, games, workshop methods, etc.

Where am I?

This paper forms part of our set of 20 guides covering all the systems innovation content. You are in the first section on systems thinking





What is Systems Thinking?

Systems thinking is a very broad area that seeks to bring together the many different ways of thinking that are holistic in their interpretation of the world. Holistic thinking seeks to understand phenomena as intimately interconnected and comprehensible with reference to the whole system or environment they form part of. Systems thinking is:

- A synthetic modes of reasoning that looks at what emerges when we put things together rather than taking them apart.[1]
- It is nonlinear way of looking at the world that focuses on relations of interdependence and feedback dynamics.
- It is a relational paradigm interpreting things in the context of the network of relations they form part of.[3]
- It is a dynamic way of looking at the world, understanding change in terms of nonlinear feedback processes that shape system structure and outcomes.[4]

Why Systems Thinking?

The primary aim of systems thinking is to elevate our thinking from just seeing parts and linear interactions to seeing and understanding whole complex systems.[5] In our everyday way of being we largely just see events, things, actions and reactions. Rarely does our thinking extend to seeing the broader systems that those events form part of.

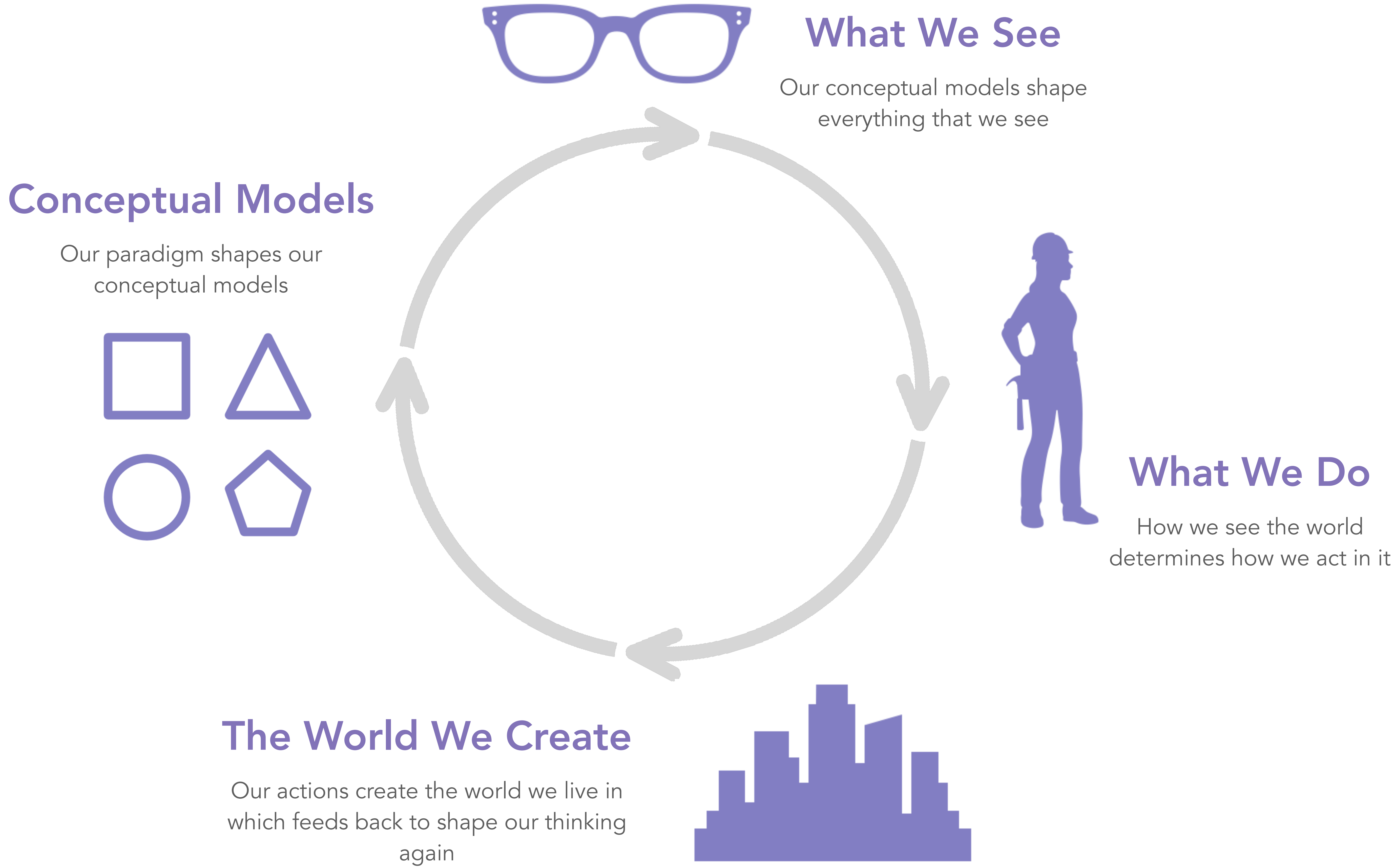
Although our thinking is often piecemeal, the reality of the world we live in is complex, messy and interconnected. Systems thinking helps to illuminate how a complex world works. If we are ever to gain a deeper more comprehensive understanding of the complexity of the real world it will involve thinking in systems.[6]

By becoming aware of the systems we form part of, the consequences of our actions within that broader context we can start to act from a different place, as individuals, as organizations, as whole societies, and from this start to overcome the unintended consequences of a partial view of the world.[7]

System Awareness

Systems thinking is, in its most generalized sense, a way of seeing the world. Before anything, it places great emphasis on the question of how do we see the world; asking us to start by being reflexive about our ways of knowing and seeing; a strong emphasis on self-awareness, awareness of one's own way of looking at the world and the way that others look at the world.[8]

At the heart of systems thinking is a recognition of our subjectivity. That how the world appears to us is not merely in some objective form, but in fact, our conceptual system structures, defines, and interprets every impression we receive. Our worldview shapes our every endeavor - whether in science, management, design or everyday life - it shapes what we do and the world we create around us. It is precisely because of this, systems thinking would posit, that we need to first understand the nature and makeup of the paradigm that we are using.


What We See
Our conceptual models shape everything that we see
Conceptual Models
Our paradigm shapes our conceptual models
What We Do
How we see the world determines how we act in it
The World We Create
Our actions create the world we live in which feeds back to shape our thinking again

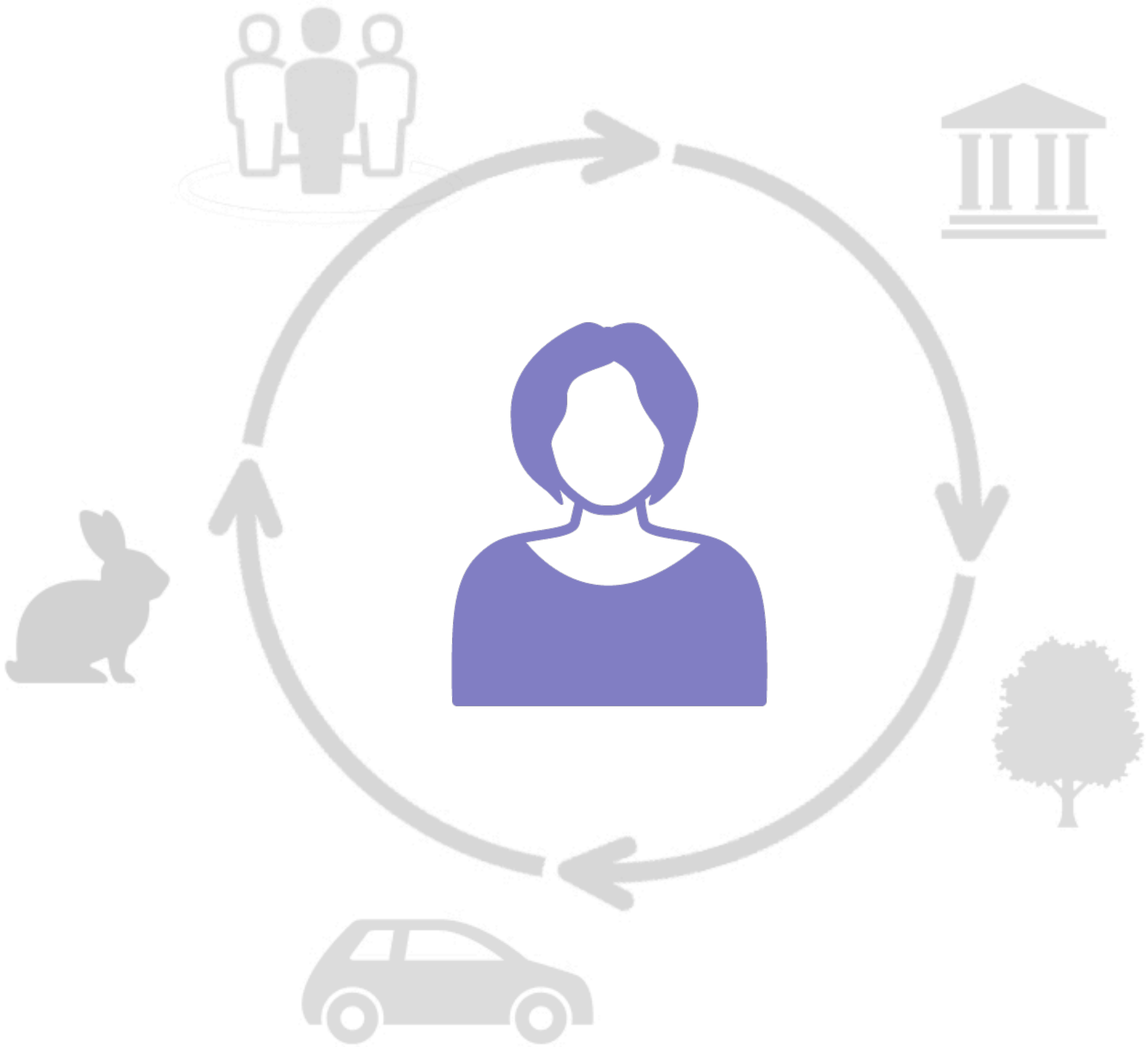
Shifting Our Thinking

If we wish to achieve some kind of systems-level innovation this is going to come from a different kind of thinking, but more than this a different kind of awareness. There is nothing more true than "the same old thinking, the same old results." It follows that the same old reductionist analytical thinking focussed on parts will surely result in the same old kind of changes to parts without systemic changes.

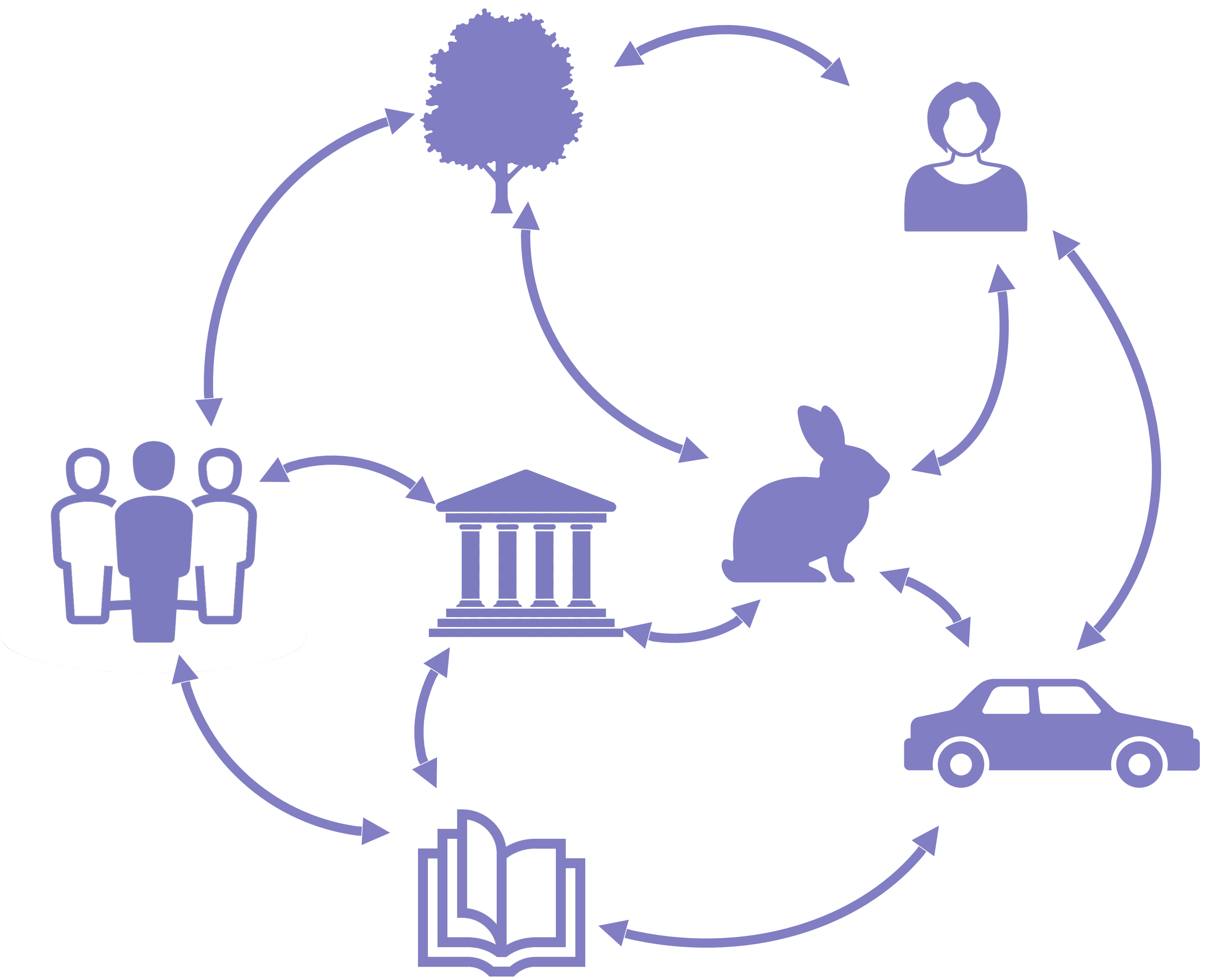
This awareness of whole systems that is the foundation for systems innovation we can call "systems awareness." It is here we will start our journey as systems innovators. The journey to systems-level innovation starts with first assessing and trying to shift our awareness and thinking from "ego-systems awareness" to "ecosystems awareness".

This icon will take you to the explainer video

Ego-system Thinking



Eco-system Thinking



Seeing Systems

Systems thinking is about trying to improve the quality of our thinking so as to not just see parts, but instead to see systems; the systems we form part of in our world. For most adults seeing systems does not come naturally, in our everyday way of being we see parts and draw a few connections that are very much centered around us and our particular place in the world. To see systems is to in some way overcome this self-centered view of the world.

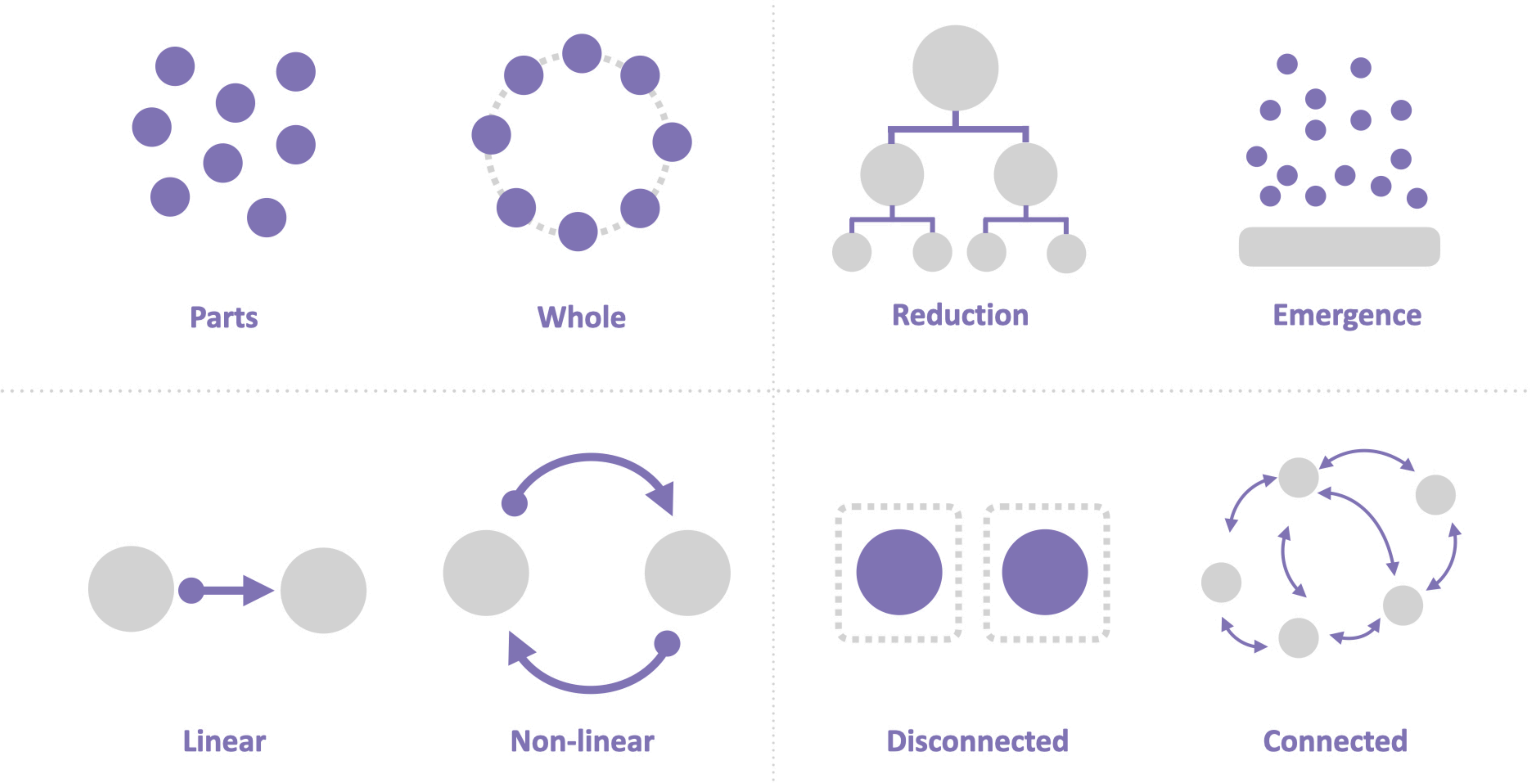
This requires us to overcome some of the limitations, flaws, and biases in our thinking resulting from our egoism so as to see the systems we form part of independent from ourselves. Calling oneself a systems thinker should be a commitment to an ongoing learning process of examining and trying to improve our thinking to become better at seeing systems. This starts with first understanding how we see the world, and the existing limitations in our reasoning.

Personal Development

Systems Change

Key Aspect of Systems Thinking

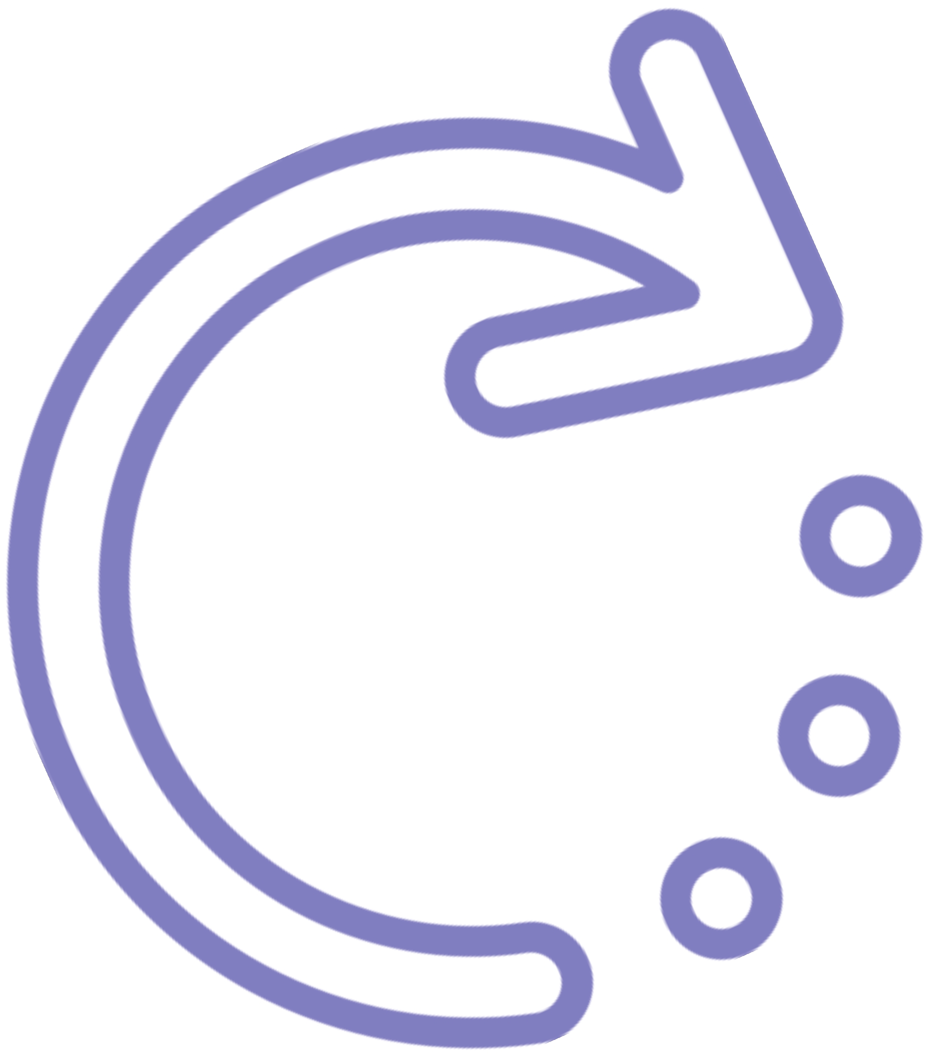
Working to shift our thinking from the left to the right of each quadrant



Holistic Thinking

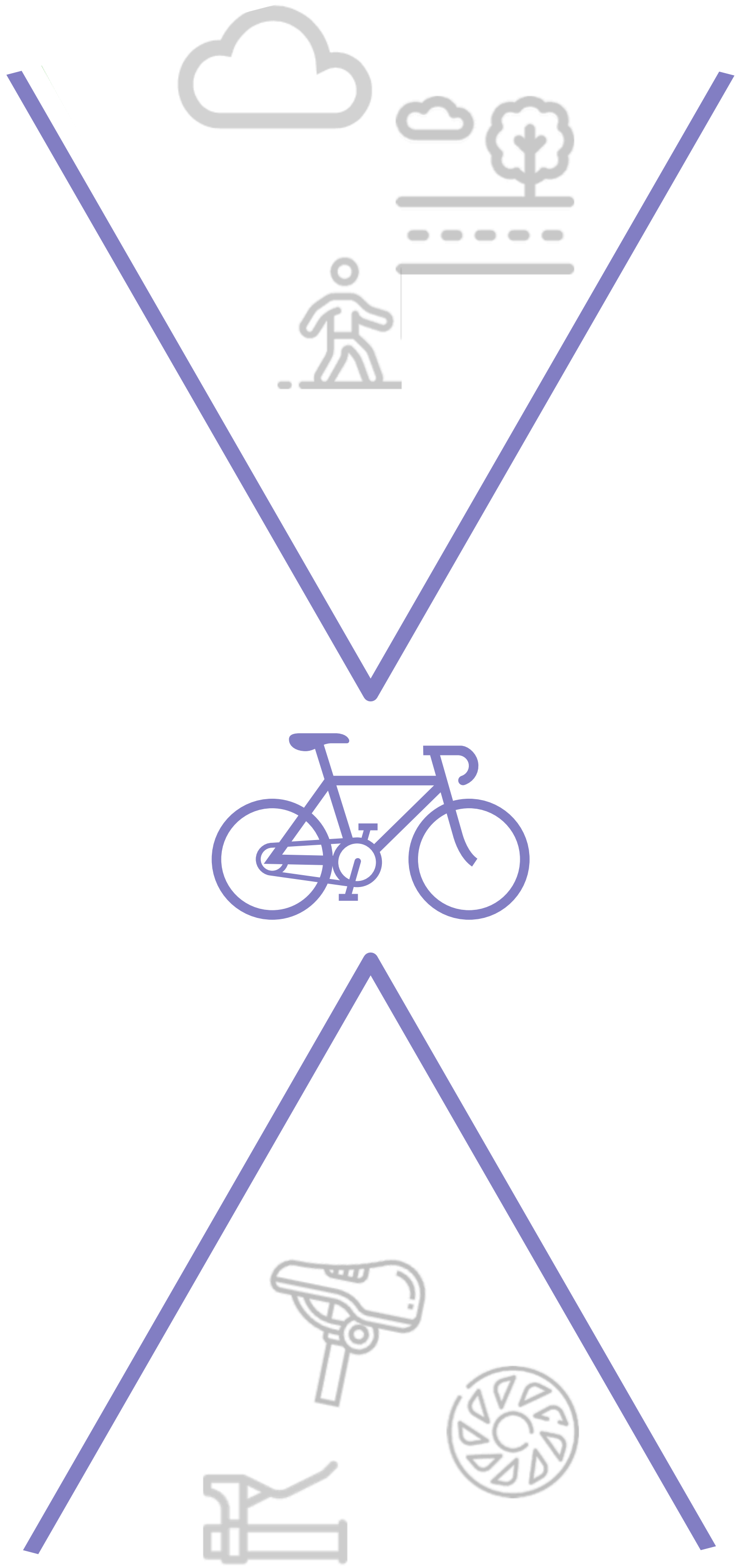
Systems thinking forms a coherent set of basic concepts that we call an "alternative" worldview. We call it alternative because what defines systems thinking is that it a more holistic paradigm and this stands in contrast to our more traditional reductionist ways of thinking.[9] Systems thinking tries to balance an analytics view of the world with a holistic one. Thus, key to becoming a systems thinker is becoming aware of the distinction between our more traditional analytical ways of reasoning and the systems approach that is holistic in nature.

Reductionism and holism form two very different views of the world and how to best interpret it. Reductionism works by breaking things down into their constituent parts and focusing on the static properties of those parts and their linear interactions. Holism does quite the opposite, trying to understand things within the context of the whole environment they form part of, their functioning within that broader context and how they are shaped by their nexus of relations.[10]



Whole

Shift the Paradigm



Parts

Holistic Thinking

Synthesizes parts to understand them with reference to the broader systems they form part of.

Is good for...

Understanding the “why"
Understanding context
Improving systems coordination
Transformational Change

Analytical Thinking

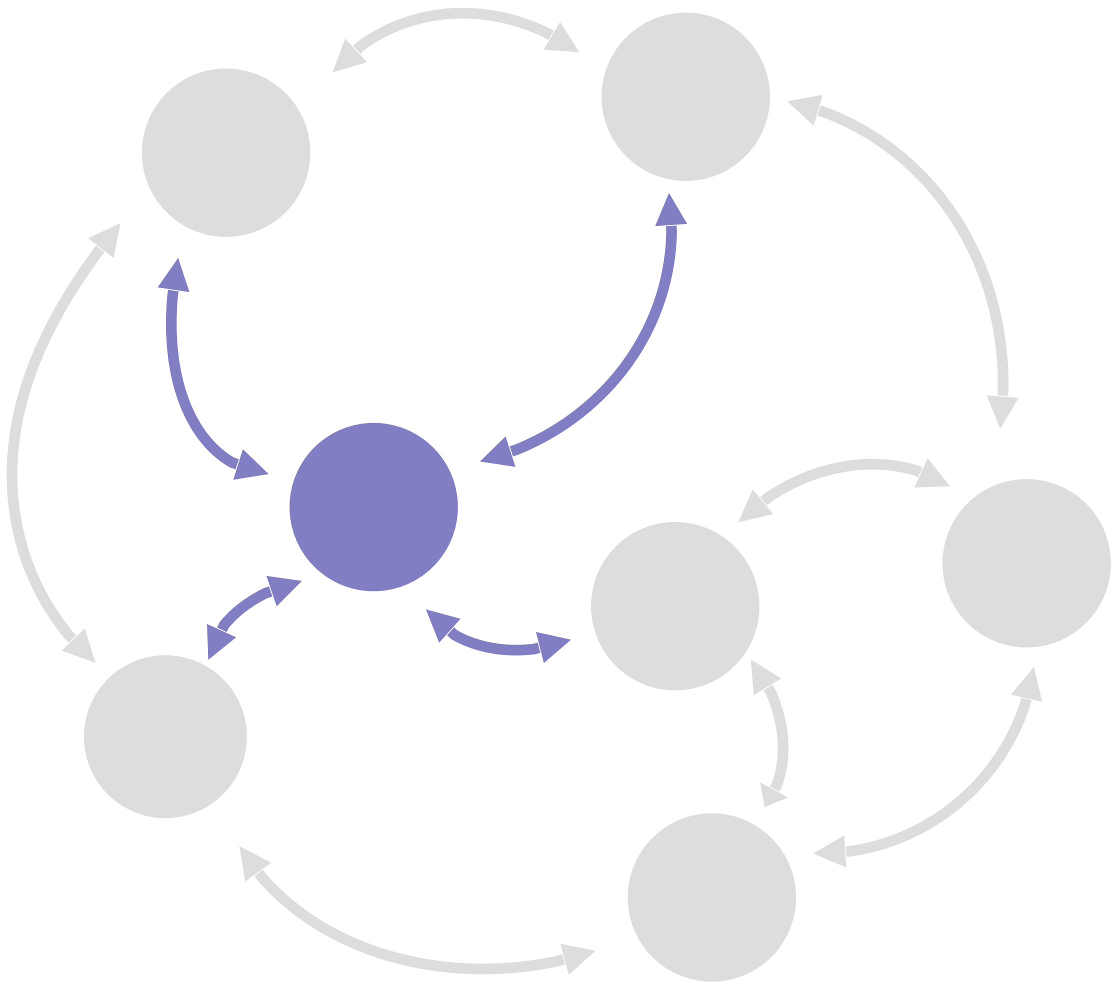
Reduces down to create an account of the whole with reference to the combination of more basic building blocks.

Is good for...

Understanding the “how"
Understanding internal workings
Increasing efficiency of parts
incremental improvements

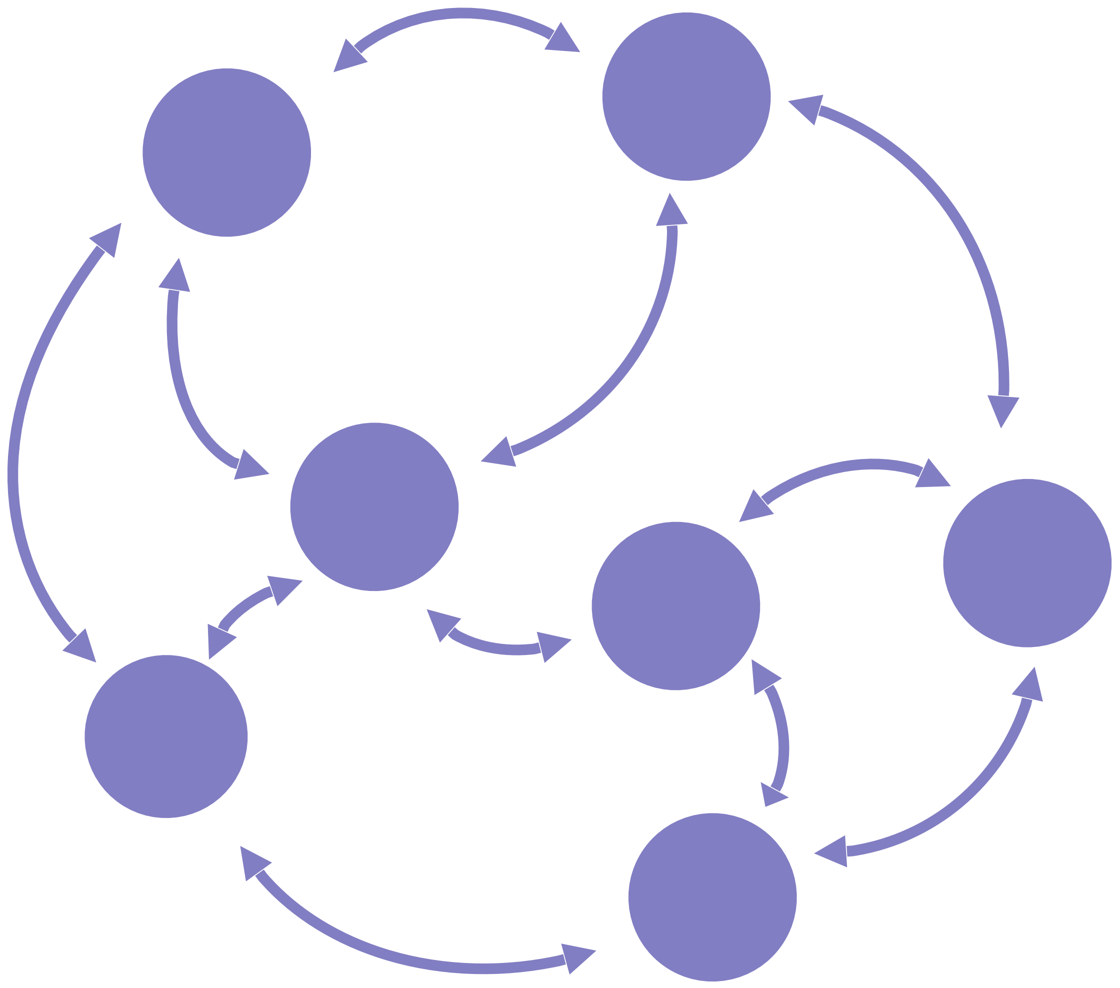
Reduction for Simple Problems

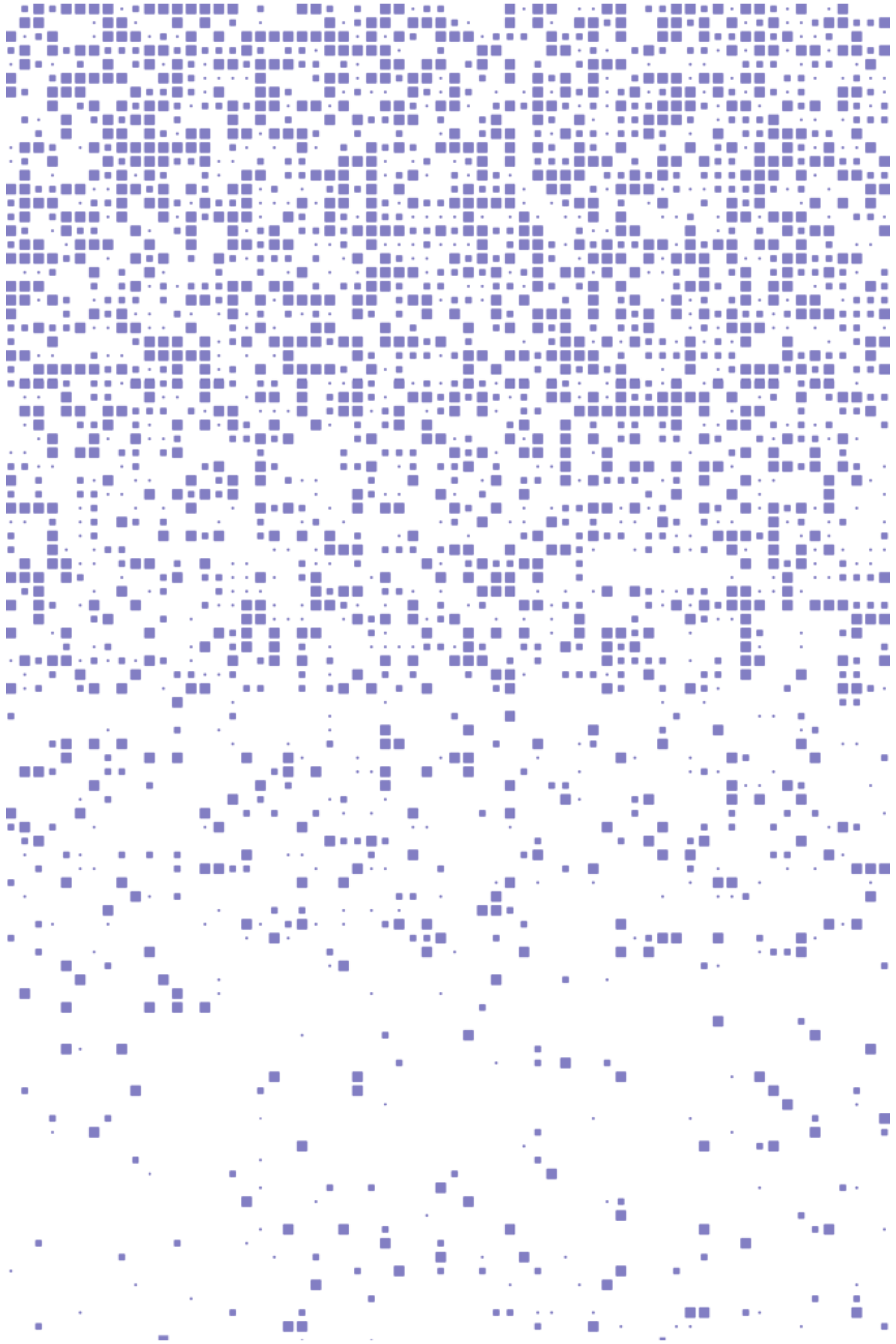
Reduce the complexity down by only focusing on the problem area, find the cause of it and change the cause to solve the problem.



Holism for Systemic Issues

Expand outwards from the perceived issue to understand the structure of the system and the overall paradigm, then influence the parts and connections in the broader network to change the pattern.





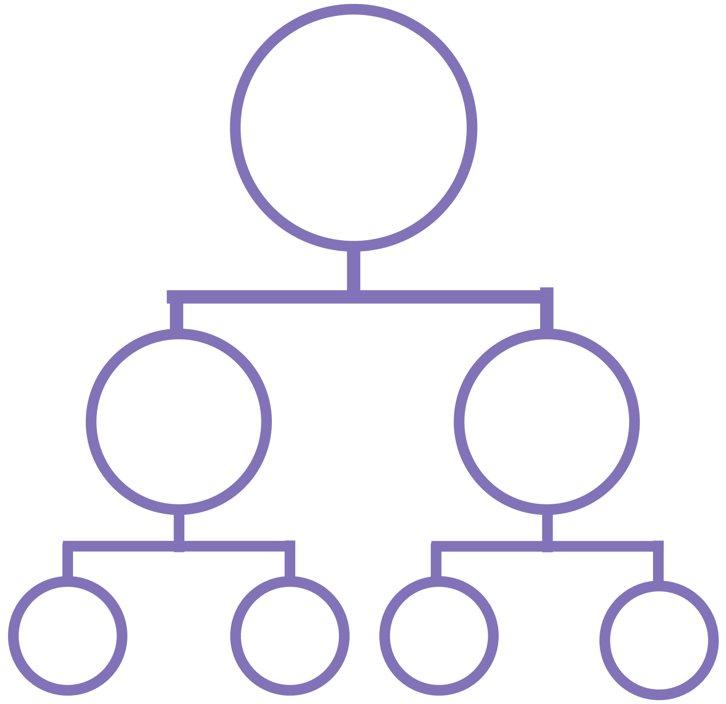
Emergence

Emergence is one of the central concepts within systems theory as it describes a universal process of becoming or creation. A process whereby novel features and properties emerge when we put elementary parts together as they interact and self-organize to create new patterns of organization.

Emergence is literally everywhere, from the evolution of the universe to the formation of traffic jams, from the development of social movements to the flocking of birds, from the cooperation of trillions of cells giving rise to the human body to the formation of hurricanes and financial crises.

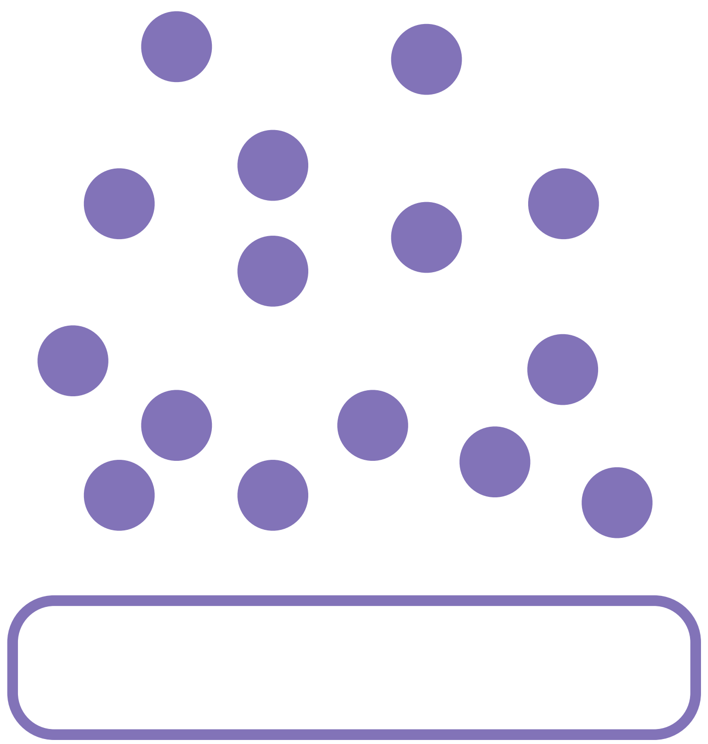
Emergence can be understood as a form of nonlinear pattern formation. Where synergies between elementary parts give rise to self-organization and the formation of a distinct pattern, that creates new, emergent levels of organization, that are driven by an evolutionary dynamic.

Emergent Thinking



Reductionism

Reductionism works to break systems down in to parts within hierarchical structures?

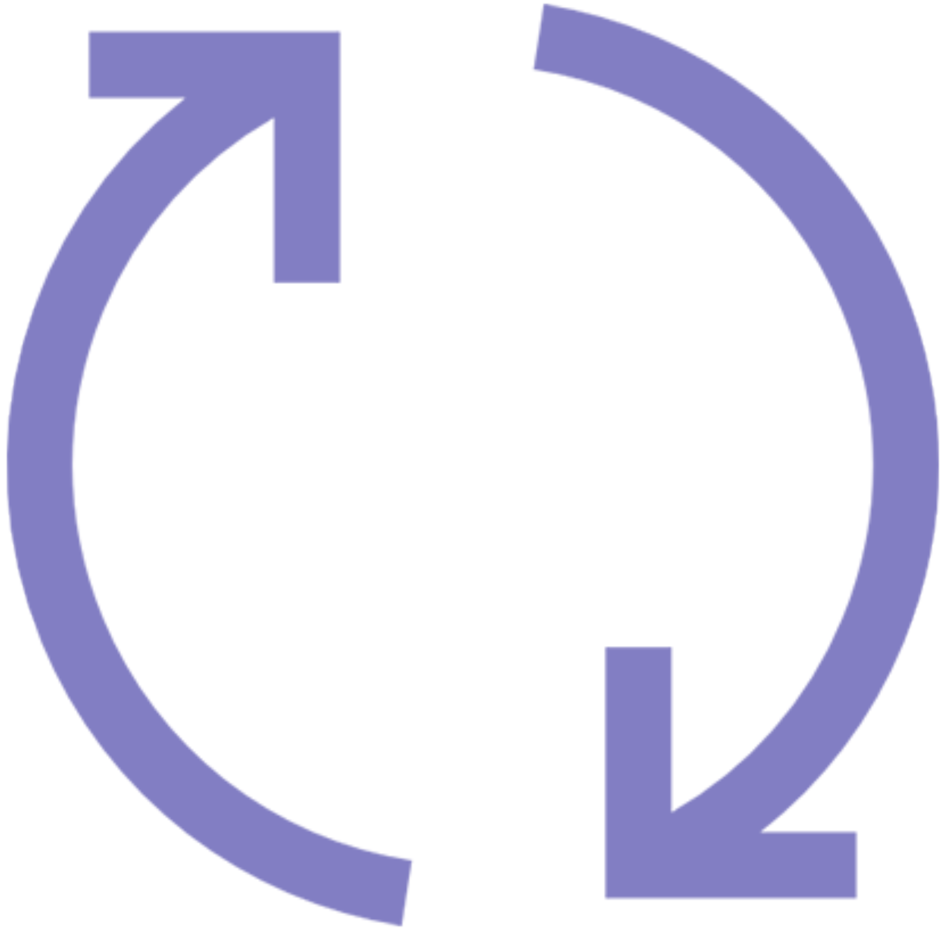


Emergence

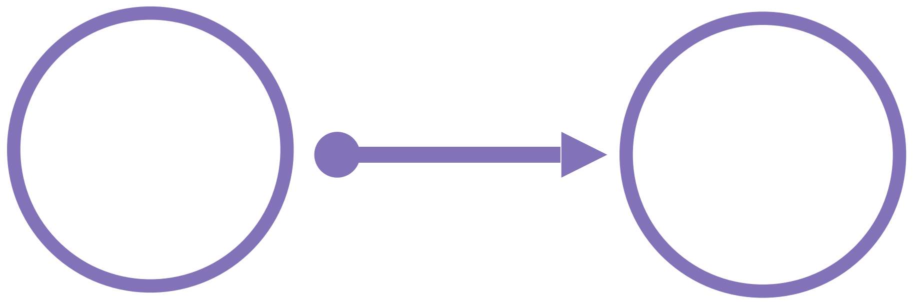
Systems thinking looks at how small parts come together in a bottom-up way to form new emergent patterns?

Nonlinear Paradigm

Systems thinking is a nonlinear way of looking at the world that emphasis the cynical interdependent nature of things that creates feedback dynamics. Feedback is at the heart of nonlinear phenomena of almost all kinds, feedback in space or over time is what generates nonlinear behavior. A feedback loop can be defined as a channel or pathway formed by an 'effect' returning to its 'cause,' and generating either more or less of the same effect. An example of this might be a dialogue between two people, what one person says now will affect what the other person will say and that will in turn feed back as the input to what the first person will say in the future.

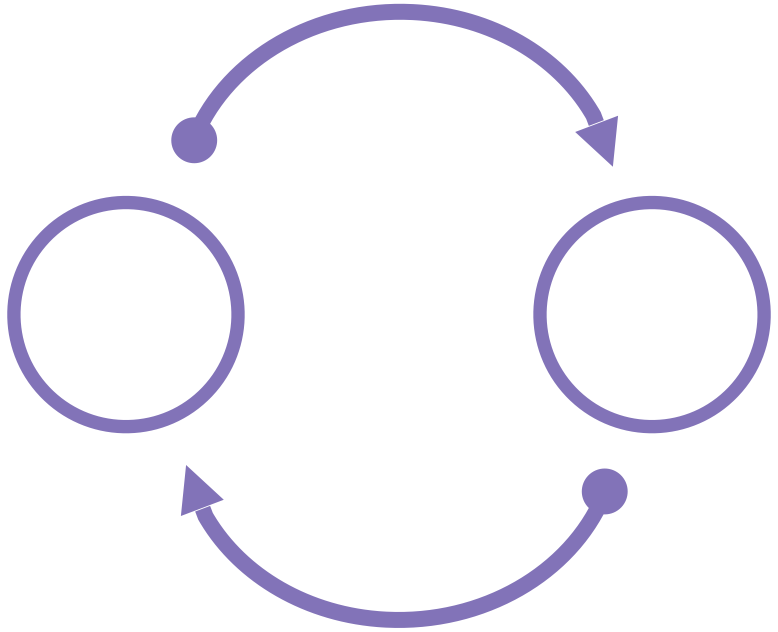


Nonlinear Thinking



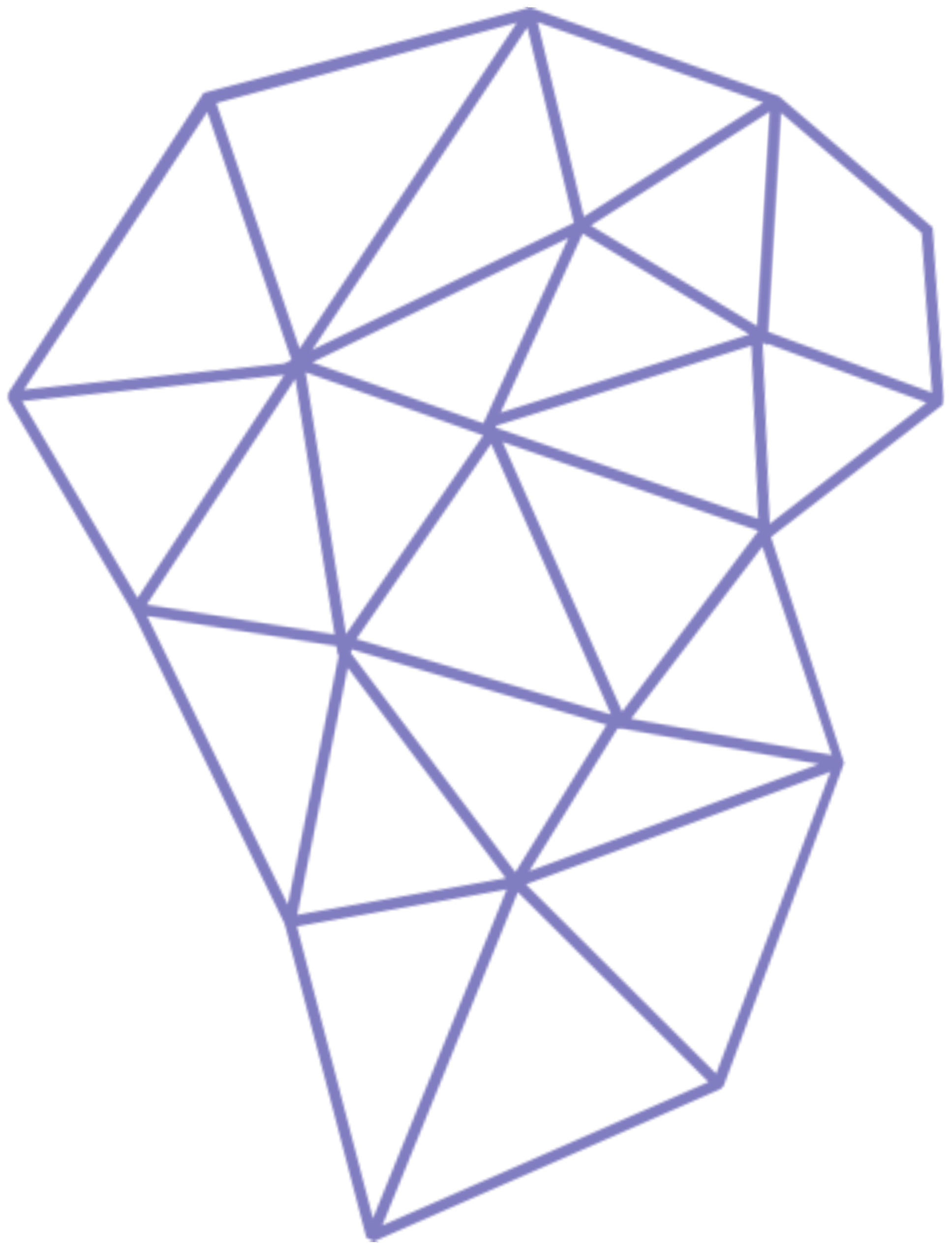
Linear Thinking

Analytical thinking is linear way of thinking looking at events as the result of simple linear interactions? For every effect or problem we search for an mediate course or set of courses



Nonlinear Thinking

Systems thinking looks are the world in a nonlinear way, think about the feedback loops in the system and cyclical causation.



Relational Paradigm

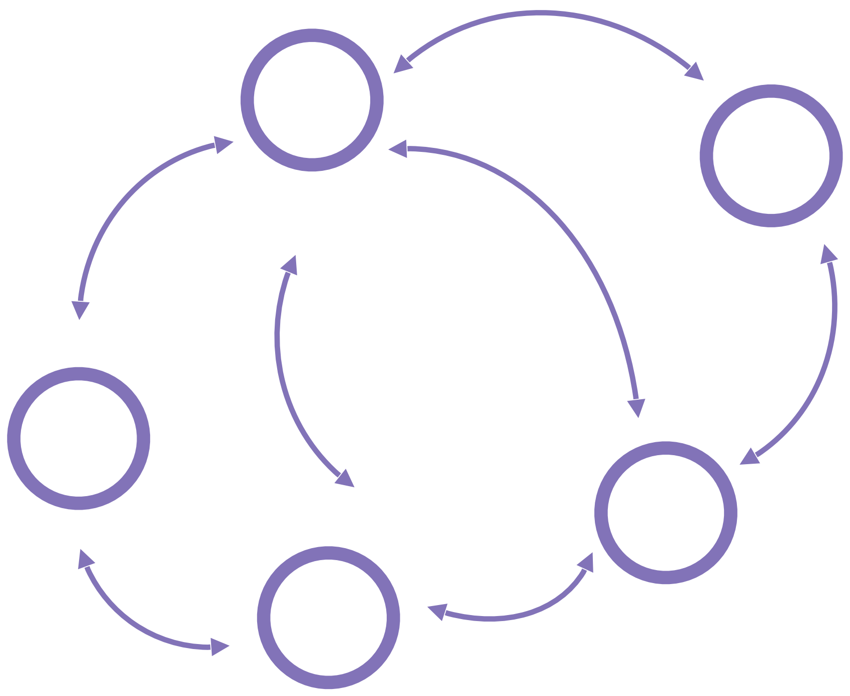
In the general sense, a system means a configuration of parts connected by a web of interdependent relationships. When talking about any system, the emphasis is typically on connections and interdependencies as the defining feature to the organization. Systems theory fundamentally rests on a relational view of the world - that is to say that the connections between parts is our primary focus rather than focusing on the parts themselves.

The relational paradigm of systems thinking emphasizes how connections, interdependencies and context shape the component parts of the system and not so much vice versa; which is the more traditional assumption. The traditional analytical paradigm is component-based. Analysis is focused on the properties of "things" in isolation and how those things cause change through direct interactions. In contrast, when looking at the world through a relational paradigm we search for how interdependencies and networks of connections shape the properties and behavior of the parts.

Relational Thinking

Disconnect Thinking

In what way do we take an isolated or disconnected view of the system?



Connected Thinking

How might we start to look at the system through a lens of interconnectivity and networks?

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